

Article

Conversion of Sewage Sludge with Combined Pyrolysis and Gasification via the Enhanced Carbon-To-X-Output Technology

Wolfgang Gebhard *, Sebastian Zant, Johannes Neidel, Andreas Apfelbacher and Robert Daschner

Fraunhofer UMSICHT, Fraunhofer Institute for Environmental, Safety, and Energy Technology, An der Maxhütte 1, 92237 Sulzbach-Rosenberg, Germany

* Correspondence: wolfgang.gebhard@umsicht.fraunhofer.de

Abstract: Sustainably produced hydrogen has the potential to substitute fossil fuels and significantly reduce CO₂ emissions. Fraunhofer UMSICHT develops a new thermochemical conversion technology to gasify ash-rich biogenic residues and waste materials that are difficult to treat with conventional gasifiers, enabling their conversion into higher-quality energy carriers such as hydrogen and syngas. Ash-rich feedstocks are difficult to convert in conventional gasification methods, as they tend to agglomerate and form slag, leading to blockages in the reactor and process disturbances. In this experimental study, hydrogen-rich syngas is produced from biogenic residual and waste materials (sewage sludge) using the Enhanced Carbon-To-X-Output (EXO) process. The EXO process is a three-stage thermochemical conversion process that consists of a combination of multi-stage gasification and a subsequent reforming step. The influence of temperature in the reforming step on the gas composition and hydrogen yield is systematically investigated. The reformer temperature of the process is gradually increased from 500 °C to 900 °C. The feedstock throughput of the pilot plant is approximately 10 kg/h. The results demonstrate that the temperature of the reforming step has a significant impact on the composition and yield of syngas as well as the hydrogen yield. By increasing the reformer temperature, the syngas yield could be enhanced. The hydrogen yield increased from 15.7 g_{H₂}/kg_{Feed} to 35.7 g_{H₂}/kg_{Feed}. The hydrogen content in the syngas significantly increased from 23.6 vol.% to 39 vol.%. The produced syngas can be effectively utilized for sustainable hydrogen production, as a feedstock for subsequent syntheses, or for power and heat generation.



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Keywords: gasification; staged gasification; biomass; hydrogen; syngas; EXO; Enhanced Carbon-To-X-Output process

1. Introduction

Hydrogen is increasingly recognized as an essential building block for a sustainable energy supply and as a key technology for the global energy transition. Its versatility as an energy carrier allows for a wide range of applications in various sectors, including industry, transport, and electricity generation. In particular, sustainably produced hydrogen has the potential to replace fossil fuels and significantly reduce CO₂ emissions. With rising global requirements for climate protection, the importance of hydrogen is rapidly increasing. According to the International Energy Agency (IEA), global demand could grow to 150 million tons annually by 2030, highlighting the need for a sustainable hydrogen economy [1]. In Germany, hydrogen is regarded a key enabler to achieve climate neutrality by 2050. National demand is estimated to reach 90–110 TWh by 2030, corresponding to a volume of 2.3–2.8 million tons. Projections see a rise of hydrogen demand to 380 TWh or ap-

proximately 10 million tons in 2050 [2]. To achieve these ambitious goals, the development and scaling of sustainable methods for hydrogen production is essential.

The use of biogenic residual materials not only provides the opportunity to recycle waste but also contributes to reducing CO₂ emissions and creates additional utilization pathways for unused residual materials [3]. In 2023, 1.63 million tons (dry substance) of sewage sludge were disposed of in Germany. Of the 1.63 million tons, 81% was thermally recycled, around 226,000 tons (14%) was used in agriculture as fertilizers or in landscaping, while approximately 85,000 tons (6%) was disposed of in other ways [4,5]. The high availability of biogenic waste materials could play a key role in thermochemical conversion processes, such as gasification technologies, for diversifying sustainable hydrogen production, especially in rural regions. Through the gasification process, biogenic materials undergo a series of thermochemical reactions, including drying, pyrolysis, oxidation, and reduction [6–10]. These processes enable the conversion of residual materials into valuable products such as syngas and hydrogen, which can then serve as feedstock for various industrial applications and products [6,9–11].

In gasification technology, there are fundamentally different types of gasifiers, distinguished by the reactor type. The various technologies differ in the type of contact between the gasification agent and biomass, the method of heat provision (allothermal or autothermal), the type of gasification agent, the pressure conditions within the reactor, the residence times in the reactor, and the number of stages (single or multi-stage processes) [7–9,11]. Conventional gasification technologies include fixed-bed reactors (counter-current and co-current), fluidized bed reactors, and entrained flow reactors. Additionally, there are advanced gasification reactors such as plasma gasification reactors, supercritical water gasification, rotary kiln gasifiers, and multi-stage gasification technologies [6–8]. An overview of the different conventional types of gasifiers is presented in Figure 1.

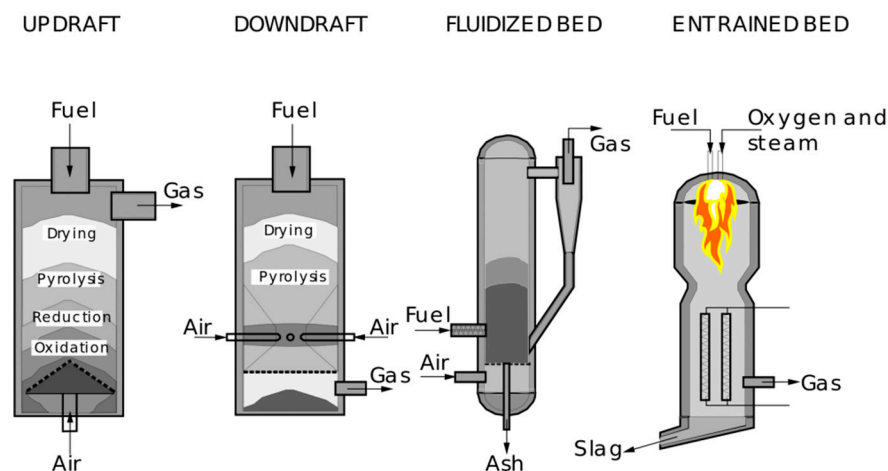


Figure 1. Fundamental types of gasifiers [12].

In the small power range (100 kW to 10 MW fuel thermal output), primarily fixed-bed gasifiers are used, in the medium power range (5–100 MW) fluidized bed gasifiers are predominately employed, and in the large power range (>100 MW) entrained flow gasifiers are utilized [7].

The newly developed fixed-bed gasification processes are mostly multi-stage systems. In some cases, fluidized bed and entrained flow gasification reactors are also integrated into multi-stage processes. These processes share the characteristic that, through technically complex two-stage process management, the disadvantages of single-stage processes are compensated [7], and synergistic interactions between chemical reactions, intermediates, and target products are maximized [9]. In multi-stage processes, the complex phases of

thermochemical conversion steps are disentangled and spatially separated to optimize process conditions and intensify subprocesses. This offers the advantage of adapting the individual conversion zones to the specific physico-chemical requirements of the feedstock. Consequently, product gasses with lower tar contents, more adjustable gas properties, higher conversion efficiencies, and consequently higher process efficiencies are produced [6,7,9].

Biomass with an ash content of less than 2% can, for example, be efficiently converted in a fixed bed counterflow gasifier. However, gasifying ash-rich feedstocks with an ash content of more than 10% is extremely challenging [13]. Due to the high ash content, these biomasses tend to form slags [13,14]. Ash particles begin to agglomerate, which leads to bridge formation and blockages in the reactor, and prevents stable process control [14,15]. If the ash content of the biomass exceeds a critical value, the proportion of alkali metals and silicon increases. At the same time, the ash melting temperature decreases. This raises the risk of sintering and the formation of agglomerates of ash particles on the surfaces of the material. These agglomerates can lead to blockages and increase the risk of slag formation [16]. Several studies have shown that ash-rich material begins to agglomerate even below the calculated ash sintering temperature [15,17,18]. Interactions of the alkaline earth metals (AAEMs) and the minerals contained in the coal lead to the formation of low-melting eutectics and consequently reduce the ash sintering temperature and ash melting temperature [15,19,20].

The newly developed and patented Enhanced Carbon-To-X-Output process (EXO process) by Fraunhofer UMSICHT is a multi-stage gasification process that enables the conversion of ash-rich and hard-to-gasify biogenic residual and waste materials. The aim of this work is to produce hydrogen-rich syngas from sewage sludge using the EXO process. The focus of this study is on varying the temperature of the reforming unit and examining the influence of temperature on gas composition and hydrogen yield.

2. Materials and Methods

2.1. Feedstock

In this study, pre-treated sewage sludge is used as the feedstock for the experiments. The sludge comes primarily from municipal wastewater and, to a lesser extent, from industrial wastewater. It is provided, dried, and pelletized by the company SET GmbH & Co. KG., Augsburg, Bavaria, Germany. The pellets are of a dark black color, which is characteristic of dried sewage sludge (Figure 2), and exhibit high mechanical strength.



Figure 2. Feedstock sewage sludge.

2.2. Experimental Setup

The pilot plant at the facility of Fraunhofer UMSICHT in Sulzbach-Rosenberg (Figure 3) has a feedstock throughput of approximately 10 kg/h. The staged allotherm process, which consists of a combination of a multi-stage gasification process and a downstream reforming step, facilitates the conversion of ash-rich and biogenic feedstocks into hydrogen-rich syngas. The parameters of pyrolysis and gasification can be specifically adapted to the requirements of the respective feedstock in the EXO process. This allows, in contrast to conventional gasification processes, the processing of feedstocks of low ash melting temperatures. In this regard, the solid residence time, temperature profile, and temperature level for each feedstock can be precisely regulated. A catalytic reforming unit is integrated downstream of the two-stage gasification for quality improvement of the product gas.



Figure 3. EXO pilot plant.

The pilot plant consists of six main components: pneumatic feed system with a double chamber lock, a horizontal screw reactor (roaster), a fixed-bed countercurrent gasifier, an ash discharge screw, a catalytic reforming reactor (reformer), and a gas treatment system, which includes a cooler and an electrostatic precipitator (ESP). The process flow diagram of the pilot plant is illustrated in Figure 4.

The material is fed into the plant using a pneumatic feed system and a subsequent rotary valve. In the first step of the EXO process, the intermediate pyrolysis of the feedstock takes place in an externally electrically heated screw reactor in the absence of oxygen. The average solid residence time in the reactor is 15 to 20 min. In the second step of the EXO process, the carbonized material produced in the roaster is gasified with a mixture of air and steam as an oxidizing agent in a fixed-bed counterflow gasifier. The resulting ash is discharged periodically into an ash container by a water-cooled discharge screw at the lower part of the gasifier. The pyrolysis vapors and gasification gasses mix in the transition area before flowing into the reformer. In the electrically heated catalytic reforming unit (step 3), the previously generated gasses are passed over a fixed-bed reactor with an activated carbon bed. The reformed gas stream is then cooled in a cooling unit to a gas temperature of 20 °C. The vapors are condensed completely and collected in a container. Subsequently, aerosols and dust are removed by the ESP. The cleaned syngas undergoes volumetric flow measurement before being burned in an external combustion chamber for odor and

emission reduction. A sampling point is available after the volumetric flow sensor, allowing us to take gas samples in gas bags for chromatographic analysis.

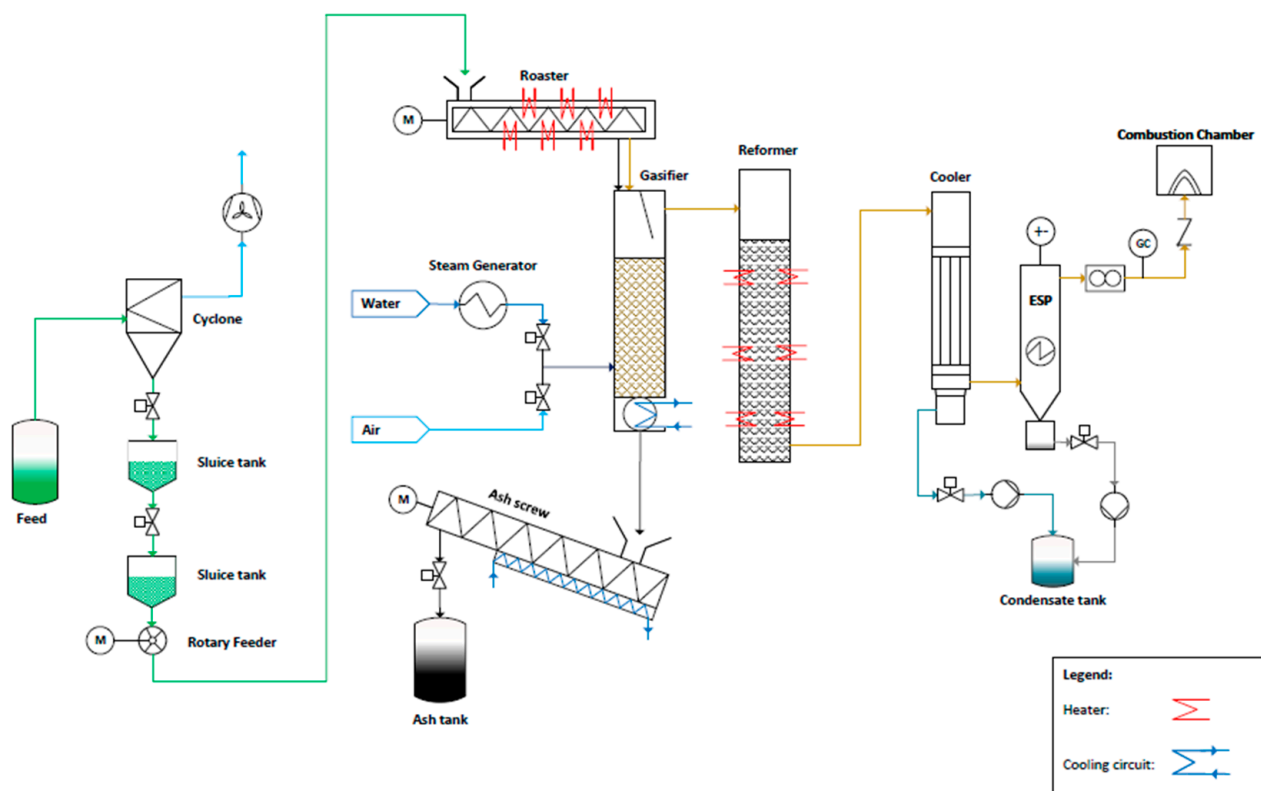


Figure 4. Process flow diagram EXO plant.

Neumann et al. demonstrate in their studies on the variation in the post-reformer temperature of the TCR[®] technology a significant correlation between the reforming temperature and the composition of the product gas [21]. The predominant chemical equilibrium reactions, such as the heterogeneous water–gas reaction, the Boudouard reaction, and the steam reforming of methane, are endothermic and therefore strongly influenced by the reforming temperature on the synthesis gas composition [7]. During the investigations conducted, the temperature of the reformer gradually varied in individual experimental campaigns from 500 °C to 900 °C. The investigated temperature range was limited to these values because, at temperatures below 500 °C, the reaction equilibrium of the reactions occurring in the reformer increasingly shifts toward the side of the reactants, such as CO₂, H₂O, and CH₄ [7]. The focus of the synthesis gas composition is on the components hydrogen and carbon monoxide. Furthermore, a temperature above 900 °C is not feasible with the current plant configuration, which is based on electric heating. At temperatures greater than 900 °C, it is expected that the previously occurring temperature-dependent effects will show the same trend due to the endothermic reactions. The operating parameters of the pyrolysis reactor and the gasifier remain unchanged during the variation. The temperature in the roaster is 450 °C, while a temperature of approximately 850 °C is maintained in the gasifier. The pyrolysis temperature was selected based on prior internal experiences with pyrolysis technologies. The gasification temperature was chosen to be below the ash melting temperature of sewage sludge. Each trial has an operating time in steady state mode of 4 h. Table 1 shows the specific operating parameters of the individual trials. Data on the syngas composition and yield, the elemental composition of the gasification ash, as well as the mass balance and the cold gas efficiency are collected based on representative sampling.

Table 1. EXO plant operating parameters of the trial series.

Parameter	Unit	Sewage Sludge					
Batch No.		2	2	2	1	1	3
Duration	h	4	4	4	4	4	4
Roaster temperature	°C	450	450	450	450	450	450
Gasifier temperature	°C	850	850	850	850	850	850
Reformer temperature	°C	500	600	700	800	850	900
Equivalence Ratio (ER)		0.13	0.13	0.12	0.13	0.13	0.12
Feedstock throughput	kg/h	10.18	10.35	11.28	11	11.25	11.27
Air supply	m ³ /h	3.77	3.79	3.77	3.76	3.76	3.74
Steam supply	m ³ /h	4.99	4.98	4.94	4.94	4.89	4.94
Nitrogen supply	m ³ /h	0.3	0.3	0.3	0.3	0.3	0.3

2.3. Analytical Methods and Measurements

After conducting the experiments, samples of the resulting products are collected and prepared for analysis. To characterize the feedstock, the gasifier ash, and the activated carbon used, an elemental analysis (CHNS), a determination of the water content, and a determination of the ash content are carried out. The analyses of the feedstock and gasifier ashes were conducted externally at the environmental analysis laboratory of Eurofins Ost GmbH. The analyses of the activated carbon were carried out internally at Fraunhofer UMSICHT in the analytical laboratory.

The elemental analysis (CHNO and S) of the feedstock was performed according to DIN EN ISO 16948:2015-09, 16994:2016-12, and 16993:2016-11. The ash content (550 °C) was measured based on DIN EN ISO 18122:2016-03. The total moisture content of the samples was measured according to DIN EN ISO 18134-2:2017-05, and the calorific value was calculated according to DIN EN ISO 18125:2017-08. Volatile components were measured according to DIN EN ISO 18123:2016-03, and fixed carbon was measured according to DIN 51734:2008-12.

The elemental analysis (CHNO and S) of the gasifier ashes was performed according to DIN EN ISO 51732:2014-07, 51724-3:2012-07, and 51733:2016-04. The ash content (815 °C) was measured based on DIN EN ISO 51719:1997-07. The total moisture content of the samples was determined according to DIN EN ISO 51718:2002-06, and the calorific value was calculated according to DIN EN ISO 51900:2023-12. Volatile components were measured according to DIN EN ISO 51720:2001-03, and fixed carbon was measured according to DIN 51734:2008-12.

The elemental analysis (CHNS) of the activated carbon was conducted internally in the analytical laboratory using the Elementar vario macro cube device. The measurements were based on the standards DIN EN ISO 16948, DIN 51724-3, and DIN 51732. Measurements were performed in triplicate, and an average value was calculated. The oxygen content was subsequently determined by difference. The moisture content of the samples was determined using a Nabertherm drying oven at 105 °C based on the standard DIN 51718. The ash content of the activated carbon was determined using a Nabertherm muffle oven based on the standard DIN 51, 719 at 815 °C. The calorific value was determined with an IKA C2000 bomb calorimeter (IKA, Staufen, Germany) according to DIN 51900. The heating value was calculated according to DIN 51900.

During the constant test operation, after approximately one hour of testing, a 1 L sample of the synthesis gas was taken at the sampling point and sent to the analytical laboratory Eurofins Umwelt Ost GmbH for measurement using gas chromatography. The

permanent gasses in the synthesis gas were measured according to DIN 51872-5:1996-08. The calorific value of the product gas was calculated according to DIN 51857:1997-03.

These investigations enable a detailed assessment of the chemical composition and provide important parameters for evaluating the process.

To comparatively evaluate the individual experiments with different reforming temperatures and derive possible correlations, a comprehensive mass balance is applied. The weight of the resulting condensate is determined by using a platform scale DE 60K10DL from the company Kern (Balingen, Germany). The weight of the feedstock, gasifier ash, and activated carbon is determined with the crane scale HCD 300K-1 from the company Kern. The gas volume produced during constant operation was measured with a diaphragm gas meter RS/2001 from the company Pietro Fiorentini (Arcugnano, Italy).

To determine the mass balance, the input and output masses of the EXO plant during the four-hour test duration are compared. The input streams consist of feedstock, nitrogen, air, water vapor, and activated carbon. The output streams consist of ash, condensate, synthesis gas, and activated carbon. The activated carbon is filled once as a batch reactor with 30 kg during the experiment. It should be noted that the water contained in the activated carbon before the experiment (8.37 wt%) evaporates during the heating process, and consequently, 27.49 kg are relevant for the mass balance. The mass balance is derived from Formula (1).

$$m_{feed} + m_{nitrogen} + m_{air} + m_{steam} + m_{active\ carbon,v} = m_{ash} + m_{condensate} + m_{syngas} + m_{active\ carbon,n} \quad (1)$$

Formula (2) describes the calculation of the mass of the total input.

$$m_{input} = m_{feed} + m_{nitrogen} + m_{air} + m_{steam} + m_{active\ carbon} \quad (2)$$

To calculate the masses of the supplied nitrogen and air, the standard volumetric flows measured on the system side are converted using the standard densities of nitrogen and air. To calculate the mass of the supplied steam, the density of the steam at the measuring point must be calculated (Formula (6)). This is conducted using the individual gas constant ($R_i = 461.5 \text{ J/kg/K}$).

$$m_{nitrogen} = \rho_{nitrogen} * V_{nitrogen} \quad (3)$$

$$m_{air} = \rho_{air} * V_{air} \quad (4)$$

$$m_{steam} = \rho_{steam} * V_{steam} \quad (5)$$

$$\rho = \frac{p}{R_i * T} \quad (6)$$

The mass of the synthesis gas is determined by difference.

$$m_{syngas} = m_{input} - m_{ash} - m_{condensate} - (m_{active\ carbon,n} - m_{active\ carbon,v}) \quad (7)$$

Formulas (8)–(11) describe the calculation of the mass fractions relevant for the mass balance in relation to the total input.

$$w_{ash} = \frac{m_{ash}}{m_{input}} \quad (8)$$

$$w_{condensate} = \frac{m_{condensate}}{m_{input}} \quad (9)$$

$$w_{active\ carbon} = \frac{(m_{active\ carbon,n} - m_{active\ carbon,v})}{m_{input}} \quad (10)$$

$$w_{syngas} = \frac{m_{syngas}}{m_{input}} \quad (11)$$

In addition to the mass balance related to the total input, the mass balance related to the feedstock is calculated. For this purpose, the amounts of nitrogen and air are simplistically subtracted from the mass of the synthesis gas (Formula (12)). The added amount of water vapor is subtracted from the amount of condensate produced (Formula (13)). The feedstock-related mass fractions are calculated according to Equations (8)–(11).

$$m_{syngas, feed} = m_{syngas} - m_{nitrogen} - m_{air} \quad (12)$$

$$m_{condensate, feed} = m_{condensate} - m_{steam} \quad (13)$$

To enable a better comparison of the process with other gasification methods, the efficiency of the process for each experiment conducted is determined based on the cold gas efficiency [22].

$$\text{Cold gas efficiency}[\%] = \frac{\text{Heating value of product gas (HHV)}}{\text{Heating value of feed (HHV)}} \quad (14)$$

3. Results and Discussion

3.1. Feedstock Characterization

The results of the investigations of the feedstock are presented in Table 2 and refer to the dried condition of the feedstock sample.

Table 2. Analysis of used sewage sludge charges.

Parameter	Unit			
Feedstock		Sewage sludge		
Batch No.		1	2	3
Total water content	wt%	8.3	11.5	8.5
C	wt%	25.2	28.4	26.6
H	wt%	3.7	4.3	4.2
N	wt%	3.6	3.95	3.3
S	wt%	0.804	1.03	0.799
O *	wt%	14.8	16.1	16.2
Ash content (550 °C)	wt%	51.9	46.1	48.9
Higher Heating Value (HHV)	MJ/kg	11.0	12.2	11.9
Lower Heating Value (LHV)	MJ/kg	10.20	11.30	11.00
Volatile components	wt%	43.9	49.3	46.8
Fixed carbon	wt%	4.2	4.6	4.3

* Determined by difference.

The results of the measurements of the three used feedstock batches (Table 2) show typical values for sewage sludge, as documented by Syed-Hassan et al. in their comparison of various sewage sludges and by other authors [23–26]. Sewage sludge exhibits a very high variability in elemental composition and ash content compared to various coals or biomasses due to its specific origin, differing production conditions, and the resulting heterogeneity. The characteristics of wastewater entering a treatment plant can significantly influence the composition of the sewage sludge. Factors such as the treatment system employed in the wastewater treatment plant, environmental legislation, water reclamation requirements, and seasonal variations also play a critical role in determining the composition [26]. These minor deviations in the CHNS-O analysis are also noticeable in the batches

used. For instance, the carbon content of the batches varies between 25.2 wt% and 28.4 wt%. The investigated sewage sludge has a very high ash content ranging from 46.1 wt% to 51.9 wt%, which varies by a few percentage points within the batches. This ash content is significantly higher than that of biomass feedstocks such as wood, which has a very low ash content of about 0.4–1.1 wt% [25–27]. The fixed carbon content of the three investigated sewage sludges ranges from 4.2 wt% to 4.6 wt%, which is slightly below the average value of 7.6 wt% published by Seyed-Hassan et al. in [26]. The share of volatile components varies between 43.9 wt% and 49.3 wt%. Seyed-Hassan et al. reported an average of 48.4 wt% of volatile components for different sewage sludges in their study [26]. The higher heating value of the used sewage sludges is between 11.0 MJ/kg and 12.2 MJ/kg, which is lower than the HHV of other sewage sludges. Seyed-Hassan et al. document HHVs for various sewage sludges ranging from 11.1 to 22.1 MJ/kg in their publication [26]. Schmitt et al. published a higher heating value of 10 MJ/kg for the sewage sludge used in their review of the thermo-catalytic reforming process [25].

3.2. Ash Characterization

After the gasification step, the sewage sludge ash consists of fine and coarse pellet-like structures (Figure 5). Despite their porous nature, the pellets exhibit high stability. The diameter and length of the pellets are reduced compared to the original material, which can be attributed to the volume reduction during pyrolysis [19]. The sewage sludge ash shows a reddish-brown coloration, which was also documented by Gil-Lalaguna et al. in their studies and is due to the presence of iron oxide in the ash [28]. The analysis results that characterize the ash are presented in Table 3. The values given refer to the dry state of the sample.

Table 3. Analysis of sewage sludge gasifier ash.

Parameter	Unit	Trials					
Reformer temperature	°C	500	600	700	800	850	900
Total water content	wt%	1.4	0.7	0.7	0.7	0.4	0.3
C	wt%	6.8	6.9	4.4	3.2	4.5	7.6
H	wt%	0.3	0.2	0.4	0.2	0.2	0.3
N	wt%	0.61	0.2	0.13	0.23	0.24	0.36
S	wt%	0.46	0.27	0.03	0.16	0.16	0.35
O *	wt%	2.9	−0.3	0.4	0.1	−0.3	0.1
Volatile components	wt%	6.4	3.1	2.9	2.9	2.6	4.0
Fixed carbon	wt%	4.5	4.2	2.4	1.0	2.3	4.6
Ash content (815 °C)	wt%	89.0	92.7	94.7	96.1	95.1	91.3
Higher Heating Value (HHV)	MJ/kg	2.62	2.47	1.84	1.35	1.75	2.51
Lower Heating Value (LHV)	MJ/kg	2.55	2.42	1.75	1.31	1.7	2.45

* Determined by difference.

The lower carbon content of sewage sludge batch 1 and the higher carbon content of batch 2 influence the elemental composition of the gasification ashes. The gasification ashes from experiments at 800 °C and 850 °C using sewage sludge batch 1 show a lower carbon content of 3.2 wt% and 4.5 wt%, respectively. In contrast, the gasification ashes from experiments at 500 °C and 600 °C show a higher carbon content of 6.8 wt% and 6.9 wt%. The increase in carbon content in the gasification ash correlates with an increase of about 3.2 wt% in the feed material. Deviations in the measurements are attributed to the inhomogeneity of the feed material as well as the associated variations in the process and the challenges in representative sampling of the ash.



Figure 5. EXO sewage sludge gasifier ash.

The carbon content of the ash has a direct influence on the lower heating value of the ash. Ashes with higher carbon content achieve a heating value of about 2.4 to 2.5 MJ/kg, while ashes with lower carbon content have heating values between 1.3 and 1.7 MJ/kg. Jäger et al. published similar relationships between carbon content and heating value of the carbonized products produced during the pyrolysis of various woody feedstocks using the TCR[®] process [29]. In comparison to biochar from woody feedstocks, which have lower heating values of 25 to 30 MJ/kg, the heating value of the gasification ash is very low due to its low carbon content [29]. The low carbon content of 3.2 to 6.9 wt% in the gasification ash indicates an efficient conversion of carbon during the gasification process. The carbon conversion rate of the two-stage gasification is approximately 80%. The carbon conversion rate is calculated from the average carbon content of the feedstock batches and the gasification ashes. The nearly complete conversion of carbon contained in the pyrolysis coke is described by Kaltschmitt et al. as a primary goal of gasification [7] and occurs through a variety of chemical reactions that can primarily be divided into heterogeneous gas–solid reactions (oxidation and reduction reactions) as well as homogeneous gas-phase reactions. The high conversion rate of carbon in the gasifier is also reflected in the ash content, which rises from about 50 wt% in the feed material to about 95 wt% in the gasification ash. Migliaccio et al. report a similar ash characteristics in the gasification of sewage sludge with a fluidized bed gasifier, which has a low carbon content of 3.17 to 7.1 wt% and a high ash content of 90.0 to 95.1 wt% [30].

3.3. Active Carbon Characterization

Activated carbon is used as a reactant in the catalytic reforming unit of the EXO plant during the experimental series. Elemental composition, water content, ash content, as well as higher and lower heating value of the activated carbon are presented in Table 4. The values given refer to the dry state of the sample.

The carbon content of the activated carbon used before the experiments is 90.42 wt% and varies between 88 and 91 wt% after the experiments. There is a trend towards increasing carbon contents of the activated carbons depending on the elevated reformer temperature. The carbon content of the activated carbon after the experiment at a reformer temperature of 850 °C deviates from this trend, which can be attributed to measurement inaccuracies in the elemental analysis and inaccuracies in the challenging sampling of the activated carbon. The observed trend of increasing carbon contents can be explained by cracking reactions of the hydrocarbon chains as well as the associated adsorption and coking reactions on the surface of the activated carbon [31–33]. The high temperatures cause the breakdown

of long-chain hydrocarbons, some of which then adsorb in the pores of the activated carbon [34]. At lower temperatures, the hydrocarbon chains are not completely cracked [7], which leads to reduced adsorption on the activated carbon [35]. The steam gasification of the activated carbon, triggered by the steam present in the gasses, results in the conversion of carbon from the activated carbon [7], which leads to a decrease in carbon content of about 1.5 wt% in the experiments at temperatures below 700 °C compared to the initial state of the activated carbon. Due to the temperature-related equilibrium shift in the endothermic steam gasification of coal, it can be assumed that the carbon content in the activated carbon should further decrease with increasing temperature. Nevertheless, the present study shows a slight increase in carbon content with rising reformer temperatures. Hosokai et al. report in their publication an equilibrium between coking reactions and steam gasification that favors the coking reactions [36]. This explains the slight increase in carbon content of the activated carbon with increasing reforming temperatures.

Table 4. Analysis of the active carbon used.

Parameter	Unit	Before Trial		Trials				
Reformer temperature	°C		500	600	700	800	850	900
Total water content	wt%	8.24	0.30	0.33	0.22	0.11	0.32	0.13
C	wt%	90.42	88.07	88.98	90.59	90.72	89.37	91.10
H	wt%	0.45	0.59	0.59	0.54	0.59	0.51	0.33
N	wt%	0.42	0.76	1.51	1.47	1.06	1.24	0.90
S	wt%	0.23	0.25	0.63	0.52	0.34	0.68	0.92
O *	wt%	2.10	2.63	1.29	0.57	−1.61	0.48	−0.67
Ash content (815 °C)	wt%	6.38	7.70	7.00	6.32	8.91	7.72	7.42
Higher Heating Value (HHV)	MJ/kg	28.59	27.50	28.41	28.51	27.52	27.74	27.81
Lower Heating Value (LHV)	MJ/kg	28.31	27.57	28.29	28.40	27.39	27.63	27.74

* Determined by difference.

The elemental analysis of the used activated carbon shows an increase in nitrogen concentrations, which can be attributed to the effect of nitrogen adsorption on the porous surface of the activated carbon [37]. The sulfur concentration of the activated carbon in the respective experiments increases with rising temperature from 0.25 wt% to 0.92 wt% due to the adsorption of H₂S on the surface of the activated carbon. Several authors demonstrated that the high specific surface area and the structure of the micro- and mesopores of the activated carbon contribute to the adsorption of H₂S [38,39]. In contrast to the elemental analysis, the interpretation of the ash concentrations of the respective activated carbons is more complex, which is attributed to fluctuations in the measured values. From the measured values, a correlation between reformer temperature and ash content of the activated carbon cannot be derived.

3.4. Product Gas Characterization

The product gas essentially consists of hydrogen (H₂), carbon monoxide (CO), carbon dioxide (CO₂), methane (CH₄), and nitrogen (N₂) (Table 5). However, a complete mass balance of the gasses cannot be performed with the measurement methods used. Long-chained hydrocarbons (C_xH_y) contained in the syngas are listed as “rest”. Additionally, the measured gas flow rates of the individual experiments, the calculated gas and hydrogen yields, as well as the lower heating value are presented.

Table 5. Syngas composition, production, and hydrogen yield.

Parameter	Unit	Trials					
Reformer temperature	°C	500	600	700	800	850	900
Methane (CH ₄)	vol.%	4.4	4.6	5.9	5.2	3.5	4.9
Carbon monoxide (CO)	vol.%	8.0	10.3	10.9	10.7	13.2	16.2
Carbon dioxide (CO ₂)	vol.%	20.7	18.4	18.1	16.9	15.4	14.8
Nitrogen (N ₂)	vol.%	38.3	33.1	31.8	32.8	28.4	22.2
Hydrogen (H ₂)	vol.%	23.6	28.9	32.0	33.2	39	38.2
Hydrogen sulfide (H ₂ S)	vol.%	0.91	0.68	0.49	0.51	0.34	0.40
Rest (C _x H _y)	vol.%	4.09	4.02	0.81	0.69	0.16	3.3
Gas production	m ³ /h	7.41	8.34	8.79	10.23	11.27	11.39
Gas yield	m ³ /kg _{Feed}	0.73	0.81	0.78	0.93	1.00	1.01
Hydrogen production	g _{H2} /h	159.8	220.2	257.0	310.4	401.3	397.4
Hydrogen yield	g _{H2} /kg _{Feed}	15.7	21.3	22.8	28.2	35.7	35.3
Lower Heating Value (LHV)	MJ/m ³	5.13	6.07	6.95	6.80	7.27	7.92

The increase in temperature during the catalytic reforming step leads to a significant increase in hydrogen yield. This effect was also described by Neumann et al. when varying the reformer temperature during the conversion of sewage sludge using TCR[®] technology [40]. Hydrogen production rises from 159.8 g_{H2}/h to about 400 g_{H2}/h with the increase in reformer temperature. In terms of feedstock, the hydrogen yield increases from 15.7 g_{H2}/kg_{Feed} at a reformer temperature of 500 °C to 35.7 g_{H2}/kg_{Feed} at 850 °C. Han et al. report in their studies on the gasification of sewage sludge using a fluidized bed gasifier an increase in hydrogen yield from 17.08 g_{H2}/kg_{Feed} to 27.87 g_{H2}/kg_{Feed} with an increase in gasification temperature from 600 °C to 800 °C [41]. The syngas yield increases with rising reformer temperature from 0.73 m³/kg_{Feed} to 1.01 m³/kg_{Feed}. Consequently, higher reforming temperatures promote gas production in the EXO process. Several published studies also confirm an increase in the amount of produced syngas depending on the reforming temperature [21,40,42,43]. Table 5 shows a decrease in the H₂S volume fraction in the syngas from 0.91 vol.% to 0.4 vol.% with increasing reformer temperatures. The decrease in hydrogen sulfide content in the syngas can be attributed to the adsorption reactions of H₂S on the activated carbon described in Section 3.3. H₂S acts as an impurity in subsequent catalyst-based syntheses, as it functions as a catalyst poison [7]. Therefore, gas purification is necessary for downstream processes. Reducing the hydrogen sulfide content can significantly decrease the effort required for gas cleaning.

Figure 6 illustrates the progression of the syngas composition of the individual components in relation to the temperature profile of the Reformer. The volume fraction of carbon dioxide in the syngas decreases with increasing Reformer temperature. The methane concentration remains nearly constant in all experiments, while the proportion of hydrogen and carbon monoxide in the syngas increases with higher reforming temperatures. Neumann et al. reported similar effects regarding the concentration of hydrogen and carbon dioxide in the product gas in their studies on the variation in post-reformer temperature in TCR[®] process during the conversion of sewage sludge [21].

The carbon dioxide content in the syngas decreases from 20.7 vol.% to 14.8 vol.%. The significant reduction of carbon dioxide in the syngas with increasing reformer temperature results from the Boudouard reaction as well as the homogeneous water–gas reaction. The reaction equilibrium of the homogeneous water–gas reaction shifts with increasing temperature in favor of carbon monoxide and water [7,8,44]. The hydrogen fraction increases from 23.6 vol.% to 39.0 vol.%, while the share of carbon monoxide in the syngas rises from

8.0 vol.% to 16.2 vol.%. The increase in carbon monoxide and hydrogen in the syngas with rising temperature results from the heterogeneous water–gas reaction [7,42,44] and the steam reforming of methane. With increasing temperature, the reaction equilibrium of the heterogeneous water–gas reaction and the Boudouard reaction shifts according to Le Chatelier’s principle in favor of carbon monoxide and hydrogen [7,44]. The reaction equilibrium of steam reforming of methane and the gasification of methane with CO_2 shifts in favor of hydrogen and carbon monoxide with increasing temperature [7,44–46], which would imply that the methane content should decrease as temperatures rise. Neumann et al. document this decline in methane concentration in their studies on the variation in the post-reformer temperature of the TCR[®] technology [21]. However, the present results show a stagnating methane content of the synthesis gas at approximately 5 vol.%, indicating that the gas residence time in the reformer is insufficient. Subsequent investigations of the residence time in the catalytic reforming unit would provide information on its influence on methane content and the overall composition of the syngas. Secondary gas-phase reactions of long-chained hydrocarbons occur during the reforming step, particularly at high temperatures, which lead to cracking into carbon monoxide and hydrogen [7,8,42,47,48]. The EXO process exhibits a characteristically high nitrogen content in the product gas when gasifying with air as the oxidizing agent. The syngas produced by this method has a high H_2 content and heating value compared to other gasification processes. The conversion of sewage sludge in a fluidized bed gasifier at 850 °C with air and steam as oxidizing agents produces syngas with a hydrogen content of 15 vol.%, a carbon monoxide content of 12.2 vol.%, a carbon dioxide content of 12.5 vol.%, a nitrogen content of 55.1 vol.%, and a methane content of 2.8 vol.% as well as a heating value of 4.7 MJ/m³ [47]. Puig-Arnavat et al. report in their publication that the product gas from a commercial downdraft gasifiers for wood exhibited a hydrogen content of 12–20 vol.% and a heating value of 5–5.9 MJ/m³ [49].

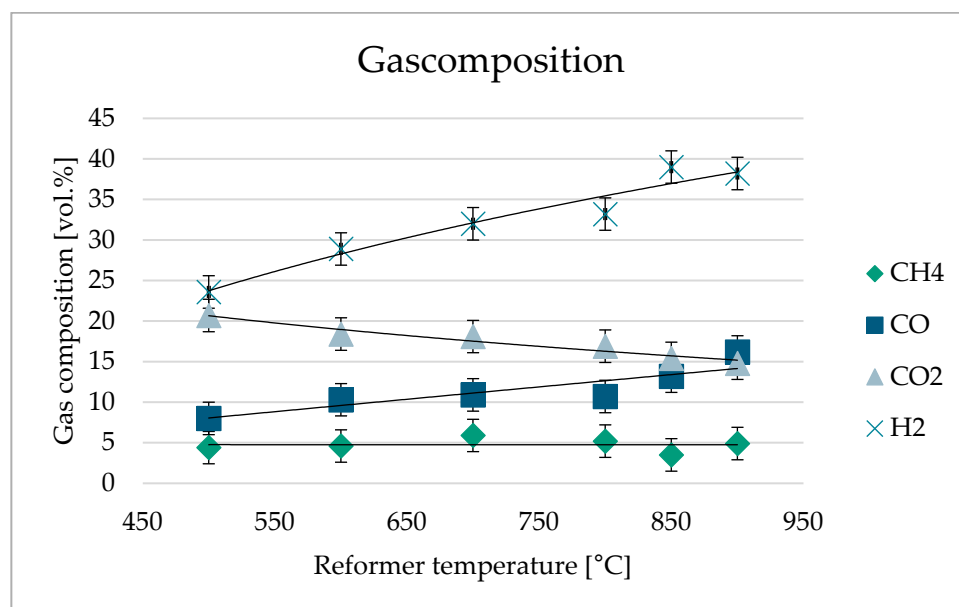


Figure 6. Gas composition of EXO syngas at different reformer temperatures.

3.5. Mass Balance

Figure 7 shows the product yield of the gasification of sewage sludge as a function of increasing reformer temperatures from 500 °C to 900 °C, based on the total input. The total input consists of feedstock, steam, air, and purge nitrogen. The yield of syngas was determined by differences.

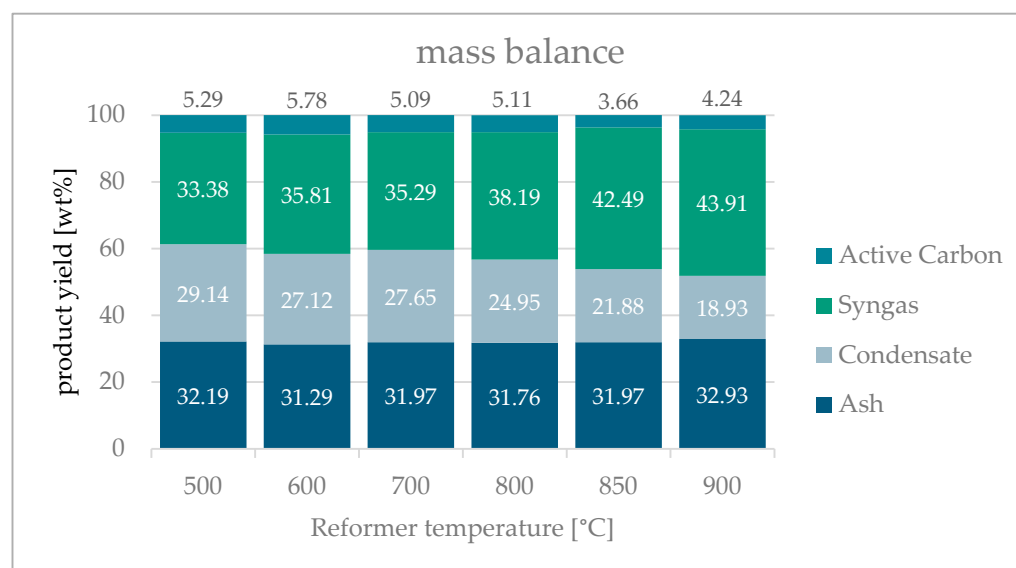


Figure 7. Mass balance of the products relative to the total input.

The mass fraction of the gasification ash relative to the total input is approximately 32 wt% in each of the experiments. The proportion of gasification ash remains nearly constant across the experiments as the reformer temperature varies. Since the reformer is placed after the gasification, it cannot influence the ash yield and composition. Minor fluctuations can be attributed to the inhomogeneity of the feedstock and the resulting process variations. The mass fraction of the condensate, consisting of process water and organic compounds, decreases from 29.1 wt% to 18.9 wt%. Similarly, the mass fraction of syngas increases from 33.38 wt% to 43.91 wt%. This increase in syngas is attributed to secondary gas-phase reactions of long-chain hydrocarbons [7,29], the heterogeneous water–gas reaction [7,42,44], as well as the steam reforming of methane [44–46]. The water contained in the gas reacts in both endothermic reactions to form carbon monoxide and hydrogen. The mass balance shows an increase in the weight of the activated carbon after the experiments, which is about 4–5 wt% after each experiment. This increase is attributed to the coking reactions of long-chain hydrocarbons at high temperatures [36,50]. Additionally, it results from the adsorption of hydrogen sulfide and nitrogen [38,39], as well as the flow-induced entrainment of particulate matter from the transition area of the roaster and gasifier.

In addition to the product distribution relative to the total input, the product distribution relative to the feedstock is also of interest (Figure 8). The yield of syngas is determined by difference.

The yield of gasification ash relative to the feedstock is approximately 55 wt% in each of the experiments, which is attributed to the high ash content of 46.1 to 51.9 wt% in the feedstock. Minor observable deviations and fluctuations in the ash balance can be attributed to the inhomogeneity of the feedstock, the resulting process variations, as well as the different feedstock batches used.

With the increase in reformer temperature from 500 °C to 900 °C, the conversion of the feedstock to gas is favored. The yield of syngas increases from 8.99 wt% to 29.62 wt%. The results also show a decrease in the yield of condensate from 22.96 wt% to 6.02 wt%.

It was observed that the weight of the activated carbon increased after the experiment. The yield of activated carbon relative to the feedstock is about 7 to 10 wt% after each experiment. This increase can be attributed to the coking reactions of long-chain hydrocarbons, the adsorption of hydrogen sulfide and nitrogen, as well as the flow-induced entrainment of particulate matter from the transition area of the roaster and gasifier. Several authors

describe in their studies the adsorption of long-chain hydrocarbons contained in the gas on the surface of the activated carbon, as well as the associated cracking and coking reactions at high temperatures [36,50]. There is a loss of micropores in the activated carbon due to coking reactions [31,33,34,36]. The steam gasification of coal forms new micropores and regenerates original micropores. Nevertheless, the number of micropores decreases throughout the investigations [36]. This suggests that a balance between coking reactions and gas–solid reactions (heterogeneous water–gas reaction) prevails, which favors the coking reactions [36]. With the increase in reformer temperature, the yield of activated carbon relative to the feedstock decreases. This is due to the shift in the balance between coking reactions and the endothermic heterogeneous water–gas reaction towards the heterogeneous water–gas reaction, which still favors the coking reactions. Furthermore, the equilibrium of the Boudouard reaction shifts due to the endothermic reaction towards the product side, favoring the formation of carbon monoxide from carbon dioxide and carbon [7].

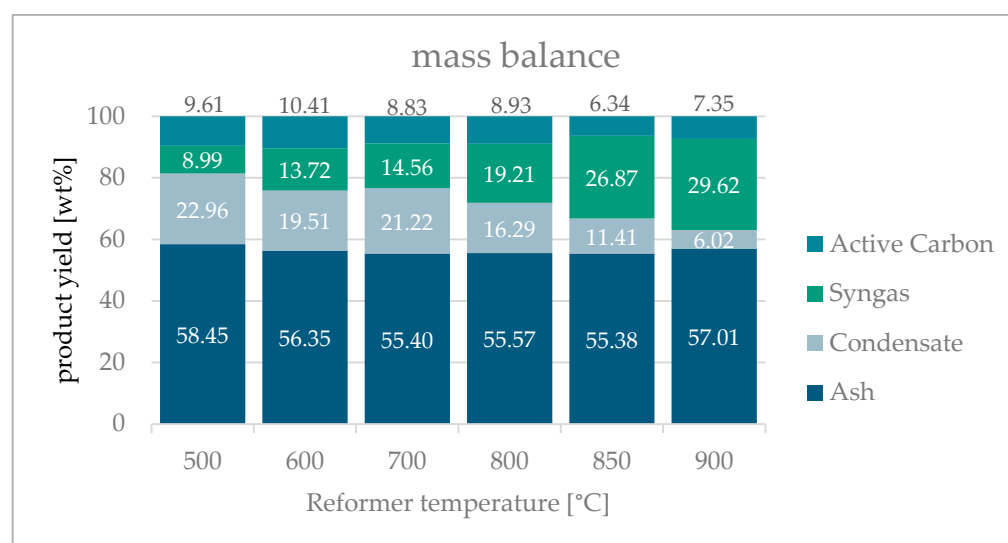


Figure 8. Mass balance of the products relative to the feedstock sewage sludge.

3.6. Cold Gas Efficiency

The cold gas efficiency defines the ratio between the energy content or power of the product gas after gasification and the energy content or power of the feedstock, and it is specifically defined for the gasification process [22,51].

The cold gas efficiency increases with rising reforming temperatures from 38.6% to 81.3% (Figure 9). This effect is attributed to the more favorable process conditions discussed in previous chapters. The correlation between the increase in the mass of syngas and the decrease in the mass of processed water is also reflected in the increase in efficiency and cold gas efficiency with higher reforming temperatures. The water present in the gas reacts in both endothermic reactions (the heterogeneous water–gas reaction and steam reforming of methane) to produce carbon monoxide and hydrogen. This conversion leads to an increase in the lower heating value of the syngas and consequently enhances the effectiveness of the process.

Compared to various other gasification processes for sewage sludge, the cold gas efficiency of the EXO plant is 81.3%, which is notably high. Several authors report a cold gas efficiency for sewage sludge gasification using downdraft gasifiers of about 60% to 65% [52,53]. Fluidized bed gasifiers achieve a similar cold gas efficiency of up to 68%, which decreases with increasing combustion air ratio (equivalence ratio) due to the rising oxidation reactions of hydrogen, carbon monoxide and methane [54]. The remarkable cold gas efficiency can be attributed to the allothermal EXO process, as the electrical

energy utilized during pyrolysis in the electrically heated screw reactor and the externally heated reformer significantly enhances the gas yield. Consequently, the cold gas efficiency increases, as the electrical energy is excluded from this calculation.

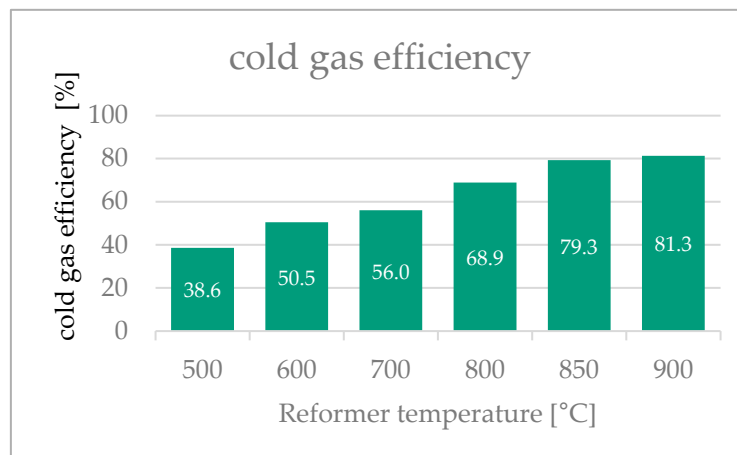


Figure 9. Cold gas efficiency of the thermo-chemical conversion of sewage sludge using the EXO process at different reformer temperatures.

4. Conclusions

This study demonstrates the effectiveness of the EXO process to utilize ash-rich feedstocks in particular sewage sludge. The results show a high carbon conversion rate of approximately 80% in the gasification unit of the EXO plant. Increasing reformer temperatures leads to a significant increase in synthesis gas production from 7.41 m³/h to 11.27 m³/h. The hydrogen yield increases from 15.7 g_{H2}/kg_{Feed} to 35.67 g_{H2}/kg_{Feed}. The hydrogen content in the synthesis gas rises with the temperature increase from 23.6 vol.% to 39 vol.%. The heating value of the EXO synthesis gas increases with rising reformer temperatures from 5.13 MJ/m³ to 7.92 MJ/m³. The results of the gas analysis indicate that the reforming temperature is crucial for the reactions occurring in the reactor. Catalytic and thermal cracking reactions of long-chain hydrocarbons as well as coking reactions on the porous surface of the activated carbon are positively influenced by higher temperatures in the reforming step. Furthermore, the reaction equilibrium of the gas–solid reactions shifts towards the product side, favoring the formation of carbon monoxide and hydrogen. Both the gas composition and the mass balance of the products are influenced by increasing reforming temperatures. With rising temperatures, there is a reduction in the condensate fraction based on the total input from 29.14 wt% to 18.93 wt%, while the synthesis gas fraction increases from 33.38 wt% to 43.91 wt%. The activated carbon experiences a mass gain due to coking reactions, which is reduced at higher temperatures due to the shift in temperature-dependent chemical equilibrium reactions. The analysis results and the mass balance of the experiments indicate that the activated carbon increasingly participates in the process as an important C-based reaction partner in the reforming unit with rising temperature. The cold gas efficiency increases with increasing reformer temperature from 38.6% to 81.3%, which is attributed to more favorable process conditions for gas-phase reactions and gas–solid reactions.

In conclusion, it can be asserted that the EXO process operates reliably, with no significant issues arising during the experimental execution. Nonetheless, the process is affected by the typical fluctuations associated with biomass due to its heterogeneous composition, presenting a challenge for effective process control. To optimize the EXO process, further investigations will be conducted to optimize operational parameters and explore different feedstocks.

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